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The Ear as a Physical Pre-processor

And why a trained ear detects AI fakes — for decades

Johann Pascher · March 2026

Speech recognition and AI audio generation have made enormous advances. Yet the human ear still surpasses digital systems in certain tasks. This has a structural cause: the ear is not a biological microphone — it is a physical computer that processes sound before the brain performs a single conscious operation. And a trained ear detects AI-generated fakes at a level no digital generator can ever fully reach.

1 The Ear Computes Before the Brain Thinks

1.1 A Long-known, Little-noticed Discrepancy

Anyone who has worked with sound signals over decades encounters a persistent observation: humans distinguish sounds, voices, and musical instruments in fractions of a second — from overlapping signals, in poor signal-to-noise conditions, in noisy environments. Digital systems still struggle with this, despite working with enormous computing power.

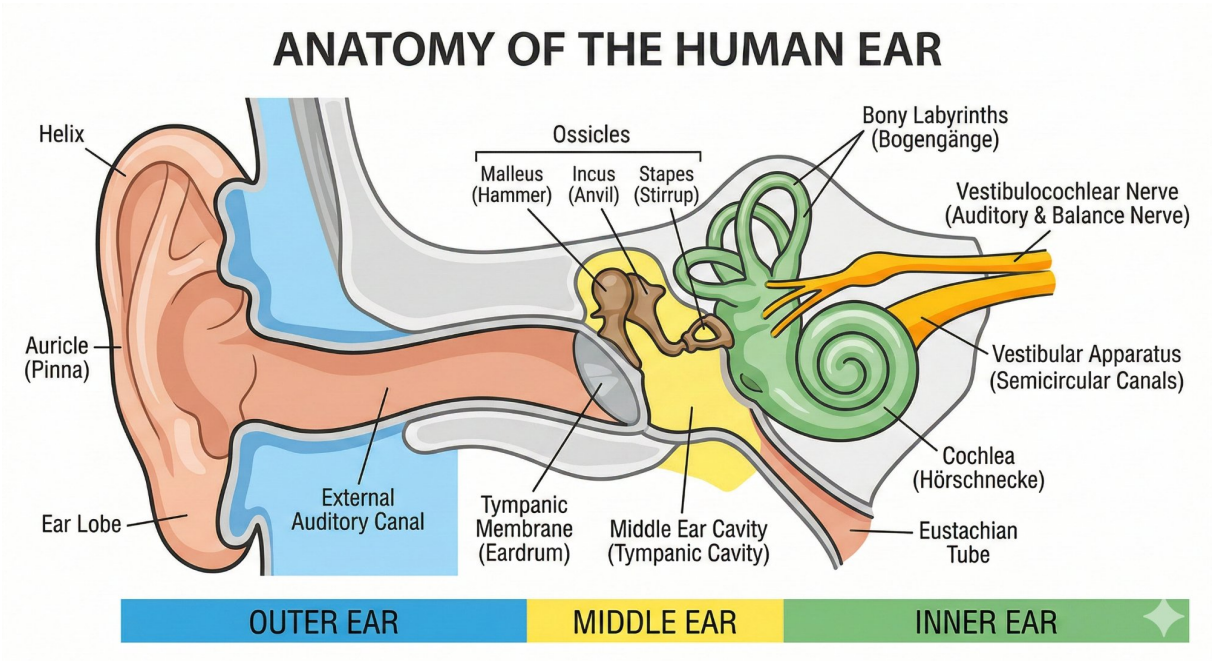
Specifically, the human ear remains superior at:

- Separating simultaneous speakers in a noisy room (cocktail-party problem)
- Identifying a single instrument within a full orchestral texture
- Detecting short transient sound events below 20 milliseconds
- Hearing far below noise levels at which digital methods fail
- Immediate voice identification even in heavily noisy signals

> The cause is not that digital systems are not yet powerful enough. The cause is structural: the ear and digital systems process sound on fundamentally different levels.

1.2 What the Ear Does Before the Brain Intervenes

The ear is not a passive recording device. It is a multi-stage analogue processing unit that fully pre-processes the sound signal in less than ten milliseconds — without a clock, without memory, without digitisation.



Anatomy of the human ear. Every stage — pinna, ear canal, eardrum, ossicles, cochlea — is an analogue pre-processor.

Pinna: Computing through shape

The geometry of the pinna filters sound as a function of frequency and direction — passively, without energy. It encodes spatial information into the frequency spectrum. A microphone cannot do this. Software can compute it afterwards, but needs several microphones and considerable processing time.

Middle ear: Matching and protection

The three ossicles (malleus, incus, stapes) match the acoustic impedance between air and inner-ear fluid — and automatically dampen excessively loud signals before the eardrum can be damaged.

Cochlea: The mechanical frequency analyser

The cochlea contains the basilar membrane: a spiral ribbon that responds to different frequencies at different positions. High tones (up to 20 kHz) excite the entrance, low tones (from 20 Hz) the far end. The sound wave decomposes into its frequency spectrum without any computation — because the mechanics of the system are built that way.

Outer hair cells: Active amplification and sharpening

The decisive element: the cochlea is not a passive system. The outer hair cells actively amplify specific frequencies through mechanical feedback. They act as a biological analogue amplifier with automatic frequency tuning — on each of the more than 3 000 frequency channels simultaneously.

The result: what reaches the brain is not a raw data transmission. It is an already prepared acoustic scene: spectrally decomposed, spatially located, dynamically compressed, selectively amplified. In less than ten milliseconds, without a clock, without a processing pause.

> *Physics computes. The brain interprets.*

1.3 The Ear and the Limits of Signal Theory

In signal processing there is a fundamental limit: the shorter the time window used to analyse a signal, the less precise the frequency information — and vice versa. This principle applies to every passive mathematical analyser that stores and then evaluates a signal.

The ear is not such an analyser. It does not store a signal and evaluate it afterwards. It responds physically in real time: the basilar membrane vibrates — continuously, without frames, without a clock. Time and frequency information arise from different

physical processes, not from the same mathematical trade-off. The ear circumvents the theoretical limit because it is a different kind of system.

Additionally, phase-locking occurs: auditory nerve fibres fire their impulses phase-coupled to the sound wave — up to approximately 4 kHz the complete temporal structure of the signal is preserved in the neural code. Frequency and time are not traded against each other — they exist simultaneously in the nerve activity.

> Provocative thesis: The ear is a physical quantum processor — not in the technical sense of qubits, but in the structural sense: a coherent wave system with active feedback that achieves information processing that passive classical systems cannot in principle reach. The limit is not computing power — it is the wrong architecture.

1.4 Animals as Living Proof

Bats locate insects to within a few centimetres through ultrasonic echolocation — in total darkness, while flying, in an environment with many other bats. The spatial resolution and processing speed of this performance would be simply impossible for a digital system solving the same task computationally at the brain size of a bat.

Dolphins analyse ultrasonic reflections so precisely that they can identify materials behind obstacles. The chimpanzee immediately recognises a voice from its social group even after years — from just a few syllables. The nervous systems of these animals are far too small for digital processing of this complexity.

> *Physics computes. The nervous system interprets.*

2 Physics is Encoded in Matter: Crystals, Cochlea, and Natural Signals

Behind the performance of the ear lies a deeper principle that goes beyond biology. The same fundamental idea is found in crystals, in photonic computing systems, and — at a fundamental physical level — in the structure of matter itself.

2.1 Crystals Compute Through Their Geometry

A crystal does not solve an eigenvalue problem — it is the solution. When light passes through a crystal, it is diffracted, filtered, reflected — according to the lattice geometry. No algorithm computes. The physics of the lattice performs the computation by existing.

The lattice constant of a crystal — the distance between atoms — is set by the atomic scale: bond lengths, atomic radii, chemical bond energy. This scale is the same as the frequency scale of the cochlea. Not a coincidence: both systems are built from matter, and matter has its fundamental scales set by the constants of physics — electron mass, fine-structure constant, Rydberg energy.

2.2 The Cochlea and the Logarithmic Scale of Nature

The cochlea analyses frequencies on a logarithmic scale: each octave (doubling of frequency) occupies approximately equal space on the basilar membrane. This is not a biological accident — it is the optimal design for the statistics of natural sound sources.

Natural sounds — speech, music, ambient sounds from wind, water, animal calls — statistically have equal energy per octave (so-called pink noise). The logarithmic cochlea scale is precisely optimised for this statistics: each frequency channel receives equal signal energy. The cochlea is calibrated to the statistics of the natural world because it is itself built from that natural world — from atomic matter with its physical constants.

2.3 Hybrid Photonic Systems: the Same Principle in Technology

The insight that physics can compute is beginning to take shape in technology. So-called Coherent Ising Machines (CIM) solve combinatorial optimisation problems by having optical oscillators physically relax into a minimum-energy state — that is the solution to the problem. A digital computer then merely reads out the result.

Cochlea	Coherent Ising Machine	Common principle
Mechanical resonance of basilar membrane	Optical resonators	Physics relaxes into solution
Active feedback by hair cells	Phase-sensitive amplifier	Active energy injection

Cochlea	Coherent Ising Machine	Common principle
Neural spike: analogue → spike	Measurement: analogue → binary	Digital post-processing
Brain interprets pre-structured data	Computer reads relaxed state	Hybrid: physics + digital

Structural parallels between cochlea and Coherent Ising Machine.

The common principle: A physical system with suitable structure relaxes into a state that solves a difficult problem. The processing is the physics — not a computation running afterwards on measurement data.

Mathematics is encoded in the geometry of matter. Physics realises this geometry. Computing is propagation.

3 The Trained Ear as Superior AI Detector

> Core statement: We would be well served to train our physical capabilities rather than relying exclusively on technical solutions. A trained human ear will be able to detect AI-generated fakes for decades to come — as long as we train it.

3.1 The Structural Problem of Automated Detection

AI-generated voices and videos are produced with neural networks. Automated detection systems use the same tools to identify them. This creates an arms race on the same technical level: a new generator architecture renders the old detector worthless. The detector is updated. The generator is improved.

This arms race is structurally won by the attacker: they set the direction, the defender reacts. The time lag between new fake and working detector can be months. In this time the damage is already done.

3.2•What Distinguishes Real Voices from AI Voices

A real voice arises from mechanics: air pressure from the lungs, vibration of the vocal folds, resonance of throat and mouth, radiation through lips and nose. Each of these processes is physical, non-linear, dependent on body and moment.

This mechanics leaves traces that are not fully reproducible in the digital domain:

- Micro-variations in pitch and loudness (jitter and shimmer): chaotic, produced by airflow and muscle tension, unique every time
- Room acoustics: reflections, reverberation, and absorption patterns of the physical space where speech was produced
- Transition sounds between words and syllables (coarticulation): arise from inertia and elasticity of tongue, lips, and palate
- Breath noise and airflow texture: turbulent flow, different every time
- Fine timbral changes due to daily condition, emotions, body posture

AI generators learn the statistical averages of these features. They produce probable signals — but not physically generated ones. The missing mechanics leaves systematic traces: overly regular micro-variation, missing room-acoustic consistency, over-smoothed tonal contours.

3.3 Why the Advantage Remains Stable for Decades

AI generators will improve — in the digital domain. They learn from more data, with larger models. But they will not improve at simulating real mechanics — because real mechanics is not digital. Vocal fold turbulence, room impulse response, breath-pressure

profile: these arise in the physical world and can only be approximately represented in the digital signal space.

The trained ear detects the missing physics. Not because it computes better than a digital system — but because it is itself physical. It is a physical system detecting physical signatures. No digital generator can attack this level because it operates on a different level.

Approach	Layer	Arms race?	Stability
Automated AI detector	Digital	Yes — attacker wins	Months to years
Trained ear	Physical	No — different level	Decades
Combination of both	Both	Partly	Optimal

3.4 Training: Concrete and Achievable

The trained ear is not a mystical talent. Radio journalists, sound engineers, and musicians develop it in daily work. It follows from deliberate practice:

- Active, deliberate listening: analysis instead of mere consumption
- Direct comparison: real and synthetic voices side by side
- Room-acoustic sensitisation: natural vs. artificial reverberation
- Working with unprocessed raw material
- Analysis of breathing patterns, speech rhythm, micro-vocal texture
- Ear training as in music education — but focused on authenticity rather than pitch

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Conclusion: We already possess the superior AI detector — it sits between our ears. It operates on the physical layer, is not attackable by digital countermeasures, works in real time without latency, and improves through training, not through computing power.

The question is not whether we should trust technology. The question is why we have stopped trusting our own ears.

The human ear is a physical processor operating at the atomic scale of matter. It works beyond the limits of passive digital systems, beyond the frame limits of sampling, beyond the reach of digital forgeries. Let us train it.